T07 HUMIDITY CONTROL | O (MAX : 1 PT)

Intent : Limit the growth of pathogens, reduce off-gassing and maintain thermal comfort by providing the appropriate level of humidity.

Summary : This WELL feature requires projects to maintain optimum relative humidity levels that are conducive to human health and well-being.

Issue : When air temperature is within a comfortable range for occupants, the effects of humidity on thermal comfort is generally uninfluential. [1] However, in warm temperature settings, humidity can influence the degradation of building materials, the ability of the human body to release heat through evaporation and the level of discomfort from excess moisture on the skin.[1–3] If the humidity is too high, the human body has a limited capacity to cool down through sweating. [4] Warm and humid indoor spaces are also associated with mold and fungal growth.[5] Moreover, humidity in warm spaces may promote the accumulation and growth of microbial pathogens, including bacteria and dust mites which can lead to odors and cause respiratory irritation and allergies in sensitive individuals.[5] Conversely, cold and dry spaces can lead discomfort and irritation of the airways, skin, eyes, throat and mucous membranes[6] and facilitate the spread of the influenza virus. [7,8]

Solutions : Buildings situated in climates with broad humidity ranges can maintain relative humidity within healthy and comfortable levels, by adding or removing moisture from the air.[9]

PART 1 MANAGE RELATIVE HUMIDITY (MAX : 1 PT)

For All Spaces:

Option 1: Mechanical humidity control

The following requirement is met in all regularly occupied areas, except high-humidity areas:

a. The mechanical system has the capability of maintaining relative humidity between 30% and 60% at all times by adding or removing moisture from the air.[10,11]

OR

Option 2: Humidity modeling

The following requirement is met for all regularly occupied areas, except high-humidity spaces:

a. The modeled relative humidity levels in the space are between 30% and 60% for at least 98% of all business hours of the year.

OR

Option 3: Long-term humidity data

The following requirements are met:

a. Project meets Feature T06: Thermal Comfort Monitoring.

b.

Relative humidity levels in regularly occupied areas, except high-humidity spaces, are between 30% and 60% during occupied hours.

Note : Projects undergoing recertification, which were previously awarded Feature T06, must consider all data collected since the previous (re)certification

Note :
Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

REFERENCES

1. Jing S, Li B, Tan M, Liu H. Impact of relative humidity on thermal comfort in a warm environment. Indoor Built Environ. 2013;22(4):598-607. doi:10.1177/1420326X12447614
2. Nematchoua MK, Orosa JA. Building construction materials effect in tropical wet and cold climates: A case study of office buildings in Cameroon. Case Stud Therm Eng. 2016;7:55-65. doi:10.1016/j.csite.2016.01.007
3. Petrofsky JS, Berk L, Alshammari F, et al. The interrelationship between air temperature and humidity as applied locally to the skin: The resultant response on skin temperature and blood flow with age differences. Med Sci Monit. 2012;18(4):CR201-CR208. doi:10.12659/MSM.882619
4. Allen J, Bernstein A, Cao X, et al. Building Evidence for Health. The 9 Foundations of a Healthy Building. Sch Public Heal. 2017:35.
5. Arundel A V., Sterling EM, Biggin JH, Sterling TD. Indirect health effects of relative humidity in indoor environments. Environ Health Perspect. 1986;VOL. 65(3):351-361. doi:10.1289/ehp.8665351
6. Wolkoff P, Kjærgaard SK. The dichotomy of relative humidity on indoor air quality. Environ Int. 2007;33(6):850-857. doi:10.1016/j.envint.2007.04.004
7. Mbithi JN, Springthorpe VS, Sattar SA. Effect of relative humidity and air temperature on survival of hepatitis A virus on environmental surfaces. Appl Environ Microbiol. 1991;57(5):1394-1399.
8. Casanova LM, Jeon S, Rutala WA, Weber DJ, Sobsey MD. Effects of air temperature and relative humidity on coronavirus survival on surfaces. Appl Environ Microbiol. 2010;76(9):2712-2717. doi:10.1128/AEM.02291-09
9. U.S. Environmental Protection Agency. Mold. https://www.epa.gov/mold.

10. U.S. Environmental Protection Agency. A Brief Guide to Mold, Moisture and Your Home.; 2012. https://www.epa.gov/sites/production/files/2016-10/documents/moldguide12.pdf.
11. European standards committee. PrEN 16798-1:2015 Indoor Environmental Input Parameters for Design and Assessment of Energy Performance of Buildings Addressing Indoor Air Quality, Thermal Environment, Lighting and Acoustics. Vol 3. European committee for Standardization; 2015.